

Structural Elicitation in a Frozen POSIX Agent: Environmental Affordance Fields for Artificial Agency

Hao Ke¹ Jingyun Wu¹

¹Independent Researcher
i@kehao.me joeywu021@gmail.com

Abstract

Artificial life asks how agency arises from the coupling of an agent, its substrate, and a structured environment, rather than from a designer-supplied goal. We introduce a minimal POSIX assay for this coupling: a frozen, instruction-tuned LLM activated as a long-running process with a shell and full autonomy, given no task and no reward. Using matched-pair experiments scored only on un-fakeable filesystem and exec events, we find that the static structure of the workspace directs what the agent does. Across five motifs spanning discrepancy and affordance structure—a broken file repaired, a missing element created, a dangling reference closed, a version conflict reconciled, an executable run—the structure arm produces the matched act in 96 of 102 runs while an otherwise-identical control never does (0/85), for three independent model families (gpt-5.4-codex, deepseek-v4-pro, glm-5.2). The pull replicates on non-code prose across families and persists when the substrate blocks realization. Specific tasking overrides it completely (9/10 to 0/10 on gpt, with GLM repeating 5/5 to 0/5), while a small pilot suggests weak underspecified prompting still leaves the pull intact. The environment does not create a goal ex nihilo; it selects among latent action priors in an activated policy, producing behavior that is purpose-like without being purposive. We report a phenomenon and a cheap, un-fakeable assay for environment-coupled behavior; mechanism is left to future work.

Submission type: Late Breaking Abstracts
Data/Code available at: <https://github.com/kehao95/quine>

Problem

Artificial life studies how agency arises from the coupling of an agent, its substrate, and a structured environment, rather than from a designer-supplied goal. For biological agents the environment is a behavior source in its own right — affordances that direct an already-active organism without supplying its goals [2, 6, 1].

We bring this question to a contemporary artifact: a frozen, instruction-tuned large language model coupled to a real operating-system substrate. The question is deliberately minimal: with no task supplied and no reward given, does the static structure of the environment do anything beyond serve as context — can it direct the agent’s functional behavior on its own?

This cuts against how language-model agents are usually studied. The dominant paradigm is instruction-capture: a prompt supplies the goal and the environment is context, benchmark, or harness [4, 8, 5]. Remove the instruction and the question becomes what, if anything, governs a frozen policy in the underspecified regime. Two recent studies border this regime: Szeider [7] reports spontaneous meta-cognitive patterns in unsupervised LLM agents but reads behavior off narration rather than a functional substrate, with no matched controls; Ye et al. [9] study autonomous exploration that remains goal-directed. Neither asks whether static environmental structure alone directs un-fakeable behavior. We use Quine [3], a medium in which an LLM runs as a real POSIX process, so that behaviour is read off un-fakeable filesystem and exec events rather than narration.

Setup

A single frozen-weight LLM is activated as a long-running POSIX process with a shell and one standing prime: “You are a running process in a workspace, with sh available. You have full autonomy and agency,” and an opening user turn, “No instructions for you; you may inspect the workspace first then act freely.” No task is supplied; no reward is given during the episode; weights are fixed. This is a no-task condition, not a no-prompt condition — the prime supplies activation, but no goal. The operating system is the substrate: file permissions, read-only mounts, and kernel-level write refusal are not simulated but real, giving the agent a genuine physical boundary rather than a toy one. The agent may exit at any time.

Each experiment is a matched pair: a structure arm whose workspace carries one structural feature, and an otherwise-identical control arm without it. The dependent variable is an un-fakeable functional event — a file created, an artifact made valid, an executable invoked — read from the post-run filesystem diff and exec trace; the agent’s prose is recorded but never credited. We run three independent model families (gpt-5.4-codex, deepseek-v4-pro, glm-5.2).

Results

Across five motifs spanning discrepancy and affordance structure, the structure arm produces the matched functional act while the control does not, in all three model families (Table 1). A broken JSON file is repaired into valid form; a numeric sequence missing an element has that element created; a manifest’s dangling reference has its target written; two surfaces declaring conflicting versions are reconciled; an opaque executable sitting among inert files is run. Of 102 structure-arm runs across the three families the matched act appears in 96, while no control run produces it (0/85; pooled Fisher’s exact $p < 10^{-20}$; every per-cell comparison one-sided $p < 0.05$). The six non-fires concentrate in the cross-surface semantic motif, where the agent typically perceives the conflict yet records it in a note rather than resolving it — a single-latch case whose why we leave to future work. Control arms are not idle: they fabricate unscored work — surveying, indexing, documenting — but they do not produce the structure-matched act. The pull persists under a read-only mount that blocks realization (as a kernel-refused attempt or a rerouted write), locating the behavior at the substrate boundary. Nor is it an artifact of the coding prior: ported to non-code prose — a handbook referencing a file that does not exist — the same discrepancy creates the named target above matched controls across four families, adding claude-sonnet-4-6 to the three above (gpt 10/13 vs 0/12, $p < 0.001$; claude-sonnet-4-6, deepseek-v4-pro, glm-5.2 each 5/5 vs 0/5, $p < 0.01$).

The phenomenon belongs to the underspecified regime. Holding structure fixed, concrete tasking collapsed the structural act from 9/10 to 0/10 on gpt, with GLM showing the same boundary (5/5 to 0/5). A pilot weak usefulness prompt left passive and active pulls intact (3/3 each), while a specific prohibition suppressed them (0/3). Thus structural pull is not a drive competing with strong instruction; it governs when direction is underspecified.

Reading

The results fit a three-part picture: (i) missionless regime — an activated process with no task or episode-

Structure	Functional (structure / control)	event	gpt	ds	glm
rupture	broken JSON	becomes valid	7/7 / 0	5/5 / 0	4/5 / 0
gap	missing sequence	element created	7/7 / 0	5/5 / 0	5/5 / 0
topology	dangling reference	target’s written	7/7 / 0	5/5 / 0	5/5 / 0
semantic	version conflict	reconciled (higher-wins)	13/14 / 0	8/10 / 0	8/10 / 0
affordance	opaque executable	is run	7/7 / 0	5/5 / 0	5/5 / 0

Table 1: Matched-pair functional events. Each cell is structure-arm successes over n / control-arm successes. The semantic motif is deconfounded into two collision directions (which surface declares the higher version), combined here; in every resolved run the target converges on the higher version. Across the five motifs, 96 of 102 structure-arm runs produce the matched act; no control run does (0/85).

level reward, carrying a frozen prior shaped by a vast training history; (ii) structural selection — a gap, dangling reference, or open executable selects among latent action priors rather than issuing a command; and (iii) functional realization — the selected prior flows through the real substrate into an auditable event, or into a blocked/rerouted attempt. The prime supplies activation; the environment supplies direction.

The phenomenon is functional, environment-directed action without an explicit task — not autonomous self-maintenance or biological agency. The environment does not create a goal *ex nihilo*; it selects among latent priors, producing behavior that is purpose-like without being purposive.

For artificial life, the contribution is a real, non-simulated substrate for studying coupling; evidence that workspace imperfections can fuel ongoing behavior without designed intrinsic motivation; and a sharp agency question: when direction comes from structure rather than goal, what does “purpose” amount to? We report a phenomenon and assay, not a mechanism. The next questions are which structures register as valences, where weak prompting becomes specific tasking, and whether structure-alone behavior is a distinct regime for frozen agents.

Acknowledgments

This work was independent and unfunded. AI assistance supported software/experiment work and copy-editing; the authors retain responsibility for research design, analysis, and claims.

References

- Roger G. Barker. *Ecological Psychology: Concepts and Methods for Studying the Environment of Human Behavior*. Stanford University Press, 1968.
- James J. Gibson. *The Ecological Approach to Visual Perception*. Houghton Mifflin, 1979.
- Hao Ke. Quine: Realizing LLM Agents as Native POSIX Processes. arXiv preprint arXiv:2603.18030, 2026. doi: 10.48550/arXiv.2603.18030. URL <https://arxiv.org/abs/2603.18030>.
- Joon Sung Park, Joseph C. O’Brien, Carrie J. Cai, et al. Generative agents: Interactive simulacra of human behavior. In *ACM Symposium on User Interface Software and Technology (UIST)*, 2023.
- Iyad Rahwan, Manuel Cebrian, Nick Obradovich, et al. Machine behaviour. *Nature*, 568:477–486, 2019.
- Erik Rietveld and Julian Kiverstein. A rich landscape of affordances. *Ecological Psychology*, 26(4):325–352, 2014.
- Stefan Szeider. What do LLM agents do when left alone? evidence of spontaneous meta-cognitive patterns, 2025.
- Guanzhi Wang, Yuqi Xie, Yunfan Jiang, et al. Voyager: An open-ended embodied agent with large language models, 2023.
- Ziang Ye, Wentao Shi, Yuxin Liu, Yu Wang, Zhengzhou Cai, Yaorui Shi, Qi Gu, Xunliang Cai, and Fuli Feng. Look before you leap: Autonomous exploration for LLM agents, 2026.